

Requirement Analysis Report

Document Information

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Source Document Title	10345 - Validate Demo E-commerce Checkout Process
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Project / Iteration / Area	Training Project - NJB4 / Iteration 1 / Training Project - NJB4
Work Item State	New
Assigned To	Sauradeep Mukherjee
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Application Under Test	SauceDemo (https://www.saucedemo.com)
Test Credentials Provided	standard_user / secret_sauce
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Business Objective

Enable a QA Tester to validate that the existing SauceDemo checkout process allows a logged-in customer to review their cart, enter shipping/checkout information, review and confirm order and payment details, and successfully complete a purchase — confirming that the flow is intuitive, secure, functional, and provides clear feedback to the customer at each step.

Note: the story is framed as a validation activity performed by a QA Tester against an already-existing checkout flow, not a request to build new functionality.

Requirement Summary

The story defines a four-stage checkout flow (Cart Review → Checkout Information Entry → Order Overview → Order Completion) plus an Error Handling criterion, gated behind login and a non-empty cart, with mandatory-field validation on the information-entry step. Five acceptance criteria (AC1–AC5) are documented, each already paired with the document's own "QA Validation" checklist, and five business rules (BR1–BR5) are documented.

Unlike the related SCRUM-101 checkout story for the same application, this document contains **no Technical Notes or Definition of Done section**, so it specifies no cross-browser, mobile-responsiveness, or performance testing scope. Several

other details relevant to QA scope — field validation formats, concrete security controls, tax calculation, cancel behavior on the information step, and whether payment method selection is in scope — are not specified in the source document and are logged as Open Questions rather than assumed.

Functional Requirements

ID	Requirement	Source
FR1	The system shall display all cart items with name, description, price, and quantity on the cart page.	AC1
FR2	The system shall display the total price calculation on the cart page.	AC1
FR3	The system shall provide a Continue Shopping option from the cart page.	AC1
FR4	The system shall provide a Proceed to Checkout option from the cart page.	AC1
FR5	The system shall redirect the user to the checkout information page when the "Checkout" button is clicked.	AC2
FR6	The system shall display form fields for First Name, Last Name, and Zip/Postal Code on the checkout information page.	AC2
FR7	The system shall treat all checkout information fields as mandatory.	AC2, BR1
FR8	The system shall display an error message indicating which field is required when a mandatory field is left empty and "Continue" is clicked.	AC2
FR9	The system shall redirect to the checkout overview page when "Continue" is clicked with valid checkout information.	AC3
FR10	The system shall display a summary of all items in the order on the overview page.	AC3
FR11	The system shall display payment information on the overview page.	AC3
FR12	The system shall display shipping information on the overview page.	AC3
FR13	The system shall display the subtotal on the overview page.	AC3
FR14	The system shall display the tax amount on the overview page.	AC3
FR15	The system shall display the total amount on the overview page.	AC3
FR16	The system shall provide a Cancel option from the overview page.	AC3
FR17	The system shall provide a Finish option from the overview page.	AC3
FR18	The system shall redirect to the order confirmation page when "Finish" is clicked.	AC4
FR19	The system shall display a success message confirming the order on the confirmation page.	AC4
FR20	The system shall display a "Back Home" button on the confirmation page.	AC4
FR21	The "Back Home" button shall navigate the user to the Products page.	AC4
FR22	The system shall clear the cart after successful order completion.	AC4, BR4
FR23	The system shall display appropriate validation error messages when invalid data (special characters, incomplete information, invalid input) is entered on the checkout information page.	AC5
FR24	The system shall prevent the user from proceeding until all checkout information fields are valid.	AC5
FR25	The system shall allow the user to proceed once invalid information has been corrected.	AC5

ID	Requirement	Source
FR26	The system shall restrict access to checkout to logged-in users only.	BR2
FR27	The system shall prevent proceeding to checkout when the cart is empty.	BR3
FR28	The system shall allow the user to cancel checkout at any step and return to the cart.	BR5

Non-Functional Requirements

ID	Requirement	Category	Source
NFR1	The checkout process shall be "secure" (no concrete controls specified).	Security	Story Description
NFR2	The checkout process shall be "intuitive" and provide "clear feedback" to the customer at each step.	Usability	Story Description
NFR3	The checkout process shall be "functional", i.e. the customer can successfully complete their purchase end-to-end.	Reliability / Functional Correctness	Story Description

Actors

Actor	Description	Source
QA Tester	Primary actor named in the story; performs the validation of the checkout process described by AC1-AC5 and their QA Validation checklists. Represented in execution by the <code>standard_user</code> test credential.	Story Description
Customer / Logged-in User	The end-user persona whose checkout experience is being validated; referenced throughout the GIVEN clauses of AC1-AC5 ("I am a logged-in user...").	AC1-AC5, BR2
SauceDemo Application (System)	The system under test that renders the cart, checkout, and confirmation pages and enforces validation.	Implied throughout

Acceptance Criteria

AC1: Cart Review

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GIVEN I am a logged-in user with items in my cart

WHEN I navigate to the cart page

THEN I should see all added items with their details:

- Name
- Description
- Price
- Quantity

AND I should see the total price calculation

AND I should have options to:

- Continue Shopping
- Proceed to Checkout

...

QA Validation (as documented):

- Verify that all products added to the cart are displayed.
- Verify product name, description, price, and quantity.
- Verify that the total price is calculated correctly.
- Verify Continue Shopping navigation.
- Verify Checkout navigation.

AC2: Checkout Information Entry

...

GIVEN I am on the cart page with items

WHEN I click the "Checkout" button

THEN I should be redirected to the checkout information page

AND I should see form fields for:

- First Name
- Last Name
- Zip/Postal Code

AND all fields should be mandatory

WHEN I leave any field empty and click Continue

THEN I should see an error message indicating which field is required

...

QA Validation (as documented):

- Verify that the Checkout button navigates to the correct page.
- Verify that all required fields are displayed.
- Verify mandatory-field validation.
- Verify the appropriate error message for each empty field.
- Verify that the user cannot continue when mandatory information is missing.

AC3: Order Overview

...

GIVEN I have entered valid checkout information

WHEN I click the "Continue" button

THEN I should be redirected to the checkout overview page

AND I should see a summary of all items in my order

AND I should see payment and shipping information

AND I should see:

- Subtotal
- Tax
- Total Amount

AND I should have options to:

- Cancel
- Finish the order

...

QA Validation (as documented):

- Verify successful navigation to the Checkout Overview page.
- Verify all selected products are displayed.
- Verify payment information.
- Verify shipping information.
- Verify subtotal calculation.
- Verify tax calculation.
- Verify total amount calculation.
- Verify Cancel functionality.
- Verify Finish functionality.

AC4: Order Completion

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GIVEN I am on the checkout overview page

WHEN I click the "Finish" button

THEN I should be redirected to the order confirmation page

AND I should see a success message confirming my order

AND I should see a "Back Home" button to return to the products page

...

QA Validation (as documented):

- Verify that the order is successfully completed.
- Verify navigation to the Order Confirmation page.
- Verify the success/confirmation message.
- Verify the Back Home button.
- Verify that Back Home navigates to the Products page.
- Verify that the cart is cleared after successful order completion.

AC5: Error Handling

...

GIVEN I am on the checkout information page

WHEN I enter invalid data, such as:

- Special characters
- Incomplete information
- Invalid input

THEN I should see appropriate validation error messages

AND I should not be able to proceed until all fields are valid

...

QA Validation (as documented):

- Verify validation for invalid data.
- Verify validation for incomplete information.
- Verify validation for special characters where applicable.
- Verify appropriate error messages.
- Verify that the user cannot proceed with invalid information.
- Verify that the user can proceed after correcting the invalid information.

Business Rules

ID	Rule
BR1	All checkout form fields are mandatory.
BR2	Users must be logged in to access checkout.
BR3	Cart cannot be empty when proceeding to checkout.
BR4	Order confirmation should clear the cart.
BR5	Users can cancel checkout at any step and return to the cart.

Dependencies

ID	Dependency	Notes
DEP1	User authentication / login feature	BR2 requires login before checkout access; the login flow itself is not specified in this document.
DEP2	Cart / product catalog feature	AC1 assumes items already exist in the cart; the add-to-cart flow is out of scope of this story.
DEP3	SauceDemo test environment and credentials	Validation depends on availability of https://www.saucedemo.com and the <code>standard_user</code> / <code>secret_sauce</code> account.
DEP4	Azure DevOps work item #10345 as the system of record	This document is an import of ADO work item #10345 (state: New); it has not yet been groomed/estimated, so the source of truth may still change upstream in ADO.

Assumptions

ID	Assumption	Rationale
A1	"Logged-in user" refers to authenticating with the provided <code>standard_user</code> / <code>secret_sauce</code> credentials.	These are the only credentials the document supplies.
A2	The checkout flow implemented in the SauceDemo application maps to: Cart page → Checkout: Your Information → Checkout: Overview → Checkout: Complete, matching AC1-AC4.	Standard SauceDemo behavior; not explicitly re-described step-by-step beyond the ACs.
A3	No real payment is processed; "payment information" shown on the overview page (AC3) is static/demo data rather than user-entered payment details.	The document defines no payment-entry step or payment form fields anywhere in the checkout flow, even though the Story Description mentions the customer being able to "select/confirm payment information" (see Open Question QQ5).
A4	Only one currency and no discount/coupon/promo-code logic is in scope.	Not mentioned anywhere in the story.

Risks

ID	Risk	Impact
R1	No Cancel control is documented for the checkout information page (AC2), yet BR5 states cancellation is possible "at any step."	Ambiguous scope could cause a missed requirement or an untestable acceptance criterion.
R2	No validation rules (allowed characters, length, format) are defined for First Name, Last Name, or Zip/Postal Code.	Negative/format test cases cannot be fully specified without guessing pass/fail boundaries.
R3	"Secure" checkout (Story Description) has no concrete, testable security requirement attached.	Security testing scope is undefined; real security gaps could go unverified.
R4	Tax calculation rule/rate is not documented, despite AC3 requiring the tax amount to be displayed.	Correctness of the displayed tax cannot be independently verified.
R5	The Story Description says the customer should be able to "select/confirm payment information," but AC3 only requires payment information to be displayed on the overview page — no payment-selection step or UI is defined anywhere in the flow.	Testers/automation cannot tell whether an unimplemented payment-selection step is a real requirement gap (defect) or the narrative wording is simply imprecise; risks either an under-tested requirement or a false defect report.
R6	Unlike the related SCRUM-101 checkout story for the same application, this document contains no Technical Notes or Definition of Done section, so it defines no cross-browser, mobile-responsiveness, or performance testing scope.	Non-functional and compatibility test coverage cannot be planned against any documented target for this story; scope may be inconsistently applied across sibling stories for the same feature.
R7	SauceDemo is a public demo site with intentionally seeded bugs on certain accounts (e.g., alternate usernames not referenced here).	If test data/accounts change, results may reflect seeded demo bugs rather than genuine defects — out of scope for this story since only <code>standard_user</code> is specified.
R8	The source Azure DevOps work item is in "New" state, indicating it has not been groomed, refined, or formally approved by the product owner.	Requirements and acceptance criteria analyzed here could still change before this story is finalized, which would invalidate downstream test artifacts derived from this report.

Edge Cases

- Attempting to proceed to checkout with an empty cart (BR3).

- Leaving exactly one of First Name, Last Name, or Zip/Postal Code empty (each field individually), vs. leaving all empty.
- Entering special characters, numeric-only, or oversized input into name fields (AC5).
- Entering invalid/malformed Zip/Postal Code formats.
- Correcting previously invalid checkout information and confirming the user can then proceed (AC5 QA Validation).
- Canceling checkout from the overview page and verifying cart contents are preserved (AC3, BR5).
- Canceling checkout from the information page (if such a control exists — see Open Question OQ2) and verifying cart contents are preserved.
- Verifying the cart is empty after order completion, then attempting to check out again (AC4 QA Validation, BR4).
- Multiple line items with varying quantities, verifying subtotal/tax/total arithmetic on the overview page.
- Using the browser back button after reaching the confirmation page, then attempting to re-finish the order.
- Session handling if login expires mid-checkout (not addressed in the source document — see Open Question OQ7).

Open Questions

ID	Question	Related Item
OQ1	What are the exact validation rules (allowed characters, min/max length, format) for First Name, Last Name, and Zip/Postal Code?	AC5, FR23, FR24
OQ2	Does the checkout information page (AC2) expose a Cancel control, given BR5 says cancellation is possible "at any step" but AC2 lists no Cancel option?	AC2, BR5
OQ3	What concrete security controls define a "secure" checkout process (e.g., transport encryption, no storage of payment data, session protections)?	Story Description, NFR1
OQ4	What tax calculation rule or rate should the overview page apply, and how should it be verified?	AC3, FR14
OQ5	Is payment method selection/entry actually part of this flow (as the Story Description's "select/confirm payment information" implies), or is the "payment information" on the overview page pre-populated/static demo data (as AC3 alone suggests)?	Story Description, AC3, FR11
OQ6	What is the expected behavior of the browser back/forward buttons at each checkout step?	Not documented in this story
OQ7	What happens if the user's session/login expires while mid-checkout?	BR2
OQ8	What browsers, devices, or viewports (if any) is this validation expected to cover, given this story — unlike the related SCRUM-101 story — includes no Technical Notes or Definition of Done section specifying compatibility/responsiveness scope?	Story Description, R6
OQ9	Does canceling checkout guarantee the cart is preserved unchanged, or could cancellation alter cart contents?	BR5, FR28
OQ10	Is this story purely a validation exercise against already-implemented checkout functionality, or could QA Validation findings result in new development work being requested (i.e., is fixing discovered defects in scope for this story)?	Story Description

Requirement Traceability

Item	Type	Source Section
FR1-FR4	Functional Requirement	AC1: Cart Review
FR5-FR8	Functional Requirement	AC2: Checkout Information Entry
FR9-FR17	Functional Requirement	AC3: Order Overview

Item	Type	Source Section
FR18-FR22	Functional Requirement	AC4: Order Completion
FR23-FR25	Functional Requirement	AC5: Error Handling
FR26	Functional Requirement	Business Rule 2
FR27	Functional Requirement	Business Rule 3
FR28	Functional Requirement	Business Rule 5
NFR1-NFR3	Non-Functional Requirement	Story Description
QA Tester, Customer / Logged-in User	Actor	Story Description, AC1-AC5, BR2
AC1-AC5	Acceptance Criteria	Acceptance Criteria section (incl. per-AC QA Validation checklists)
BR1-BR5	Business Rule	Business Rules section
DEP1	Dependency	BR2 (inferred)
DEP2	Dependency	AC1 (inferred)
DEP3	Dependency	Application URL / Test Credentials
DEP4	Dependency	Source metadata (Azure DevOps work item)
A1-A4	Assumption	Inferred from Application URL, Test Credentials, AC3, Story Description
R1-R8	Risk	Inferred from gaps across AC2, AC3, Story Description, and source metadata
OQ1-OQ10	Open Question	Inferred from gaps across AC2, AC3, BR5, Story Description

QA Coverage Recommendations

1. Positive-path coverage for the full checkout flow: Cart Review → Checkout Information → Order Overview → Order Completion (AC1-AC4), as a smoke/regression test.
2. Incorporate the document's own per-AC "QA Validation" checklists directly into the structured test plan and manual test cases — they already enumerate the specific assertions the story author expects (e.g., "Verify Continue Shopping navigation", "Verify tax calculation").
3. Mandatory-field validation coverage for AC2: each of First Name, Last Name, Zip/Postal Code left empty individually, in pairs, and all together.
4. Negative/format validation coverage for AC5: special characters, whitespace-only input, incomplete information, and boundary-length input — pending clarification in OQ1 before exact pass/fail rules can be finalized.
5. Positive recovery coverage for AC5: after triggering a validation error, correct the invalid field(s) and verify the user can then proceed (explicitly called out in the story's own QA Validation list).
6. Cart-state verification: total price accuracy on the cart page (AC1) and subtotal/tax/total accuracy on the overview page (AC3), across single-item and multi-item/multi-quantity carts.
7. Business-rule negative tests: attempt checkout with an empty cart (BR3) and attempt to access checkout while logged out (BR2).

8. Cancel-flow coverage: cancel from the overview page (AC3) and, pending OQ2, cancel from the information page — verify cart contents are preserved in both cases (BR5, OQ9).
9. Post-completion state check: verify the cart is cleared after order confirmation (AC4 QA Validation, BR4) and that "Back Home" (AC4) returns to the Products page correctly.
10. Flag rather than assert against the undefined "secure", "intuitive", and payment-selection qualities (NFR1, NFR2, OQ3, OQ5) during exploratory testing rather than encoding assumptions into automated checks.
11. Confirm with the product owner whether cross-browser and mobile-responsiveness coverage is in scope for this story (OQ8, R6) before excluding it from the test plan — the sibling SCRUM-101 story for the same application specifies it, but this document does not.
12. Do not write automated assertions against tax calculation values, Zip/Postal Code format limits, payment-selection UI, or Cancel destination from the information page until OQ1, OQ2, OQ4, and OQ5 are resolved with the product owner.